

Mr. Kang Bochun

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EDUCATION BACKGROUND

The Hong Kong Polytechnic University (PolyU)	01/2023-12/2026
• <i>Doctor of Philosophy</i> in Applied Physics	
University College London (UCL)	09/2021-12/2022
• <i>Master of Science</i> in Advanced Materials Science	
Queen Mary University of London (QMUL)	08/2017-07/2021
• <i>Bachelor of Engineering</i> with First-class Honours in Material Science and Engineering	
Northwestern Polytechnical University (NPU) (985 & 211 Project)	08/2017-07/2021
• <i>Bachelor of Engineering</i> in Material Science and Engineering	

PROFESSIONAL SKILLS

Process & Simulation:

- **Scalable Devices Fabrication:** Spin coating, blade coating, PVD, E-Beam, photolithography, mask design (CAD), laser scribing
- **Simulation:** Fluid dynamics optimization (Ansys Fluent CFD simulation), SCAPS-1D solar cell simulation, Silvaco TCAD

Characterization:

- **Semiconductor Carriers & Optoelectronics:** SCLC, PL, TRPL, EIS, Mott-Schottky, Admittance
- **Morphology & Microstructure:** SEM, AFM, XRD, TEM
- **Composition, Chemistry & Band Alignment:** XPS, ToF-SIMS, UPS, Raman, UV-Vis, KPFM
- **Device Performance & Reliability:** J-V Curves, EQE, SPO, ISOS Standards

RESEARCH EXPERIENCE

Work 1: Blade design aids in high-quality, scalable uniform thin-film HTL and perovskite deposition & CFD simulation

- **Highlights:** Led the fluid dynamics optimization for scalable thin-film deposition. Designed customized blade-coating architectures and conducted comprehensive CFD simulations (Ansys Fluent) to regulate hydrodynamic behaviors, achieving high-uniformity deposition of perovskite and HTL, a critical step for industrial manufacturing scale-up of perovskite solar modules (PSMs). Ultimately delivering a remarkable power conversion efficiency (PCE) of 22.1% on a 7-cell PSMs with an active area of 15.32 cm².

Work 2: Laser patterning and monolithic interconnection design for large-area PSMs

- **Highlights:** Designed and optimized the laser scribing processes (P1, P2, P3) for monolithic interconnection in large-scale perovskite solar modules, successfully minimizing the dead zone width and enhancing the geometric fill factor (GFF) to 98%. Investigated the laser-induced heat-affected zones (HAZ) and film delamination mechanisms to suppress interfacial shunt leakage, ensuring high efficiency and stability in upscaled photovoltaic devices.

Work 3: Photolithography design of micropyramid-structured PDMS anti-reflection layers for enhanced light harvesting

- **Highlights:** Developed a high-precision photolithography and soft-lithography microfabrication protocol to engineer micro-structured PDMS anti-reflection layers. Utilized AutoCAD for mask layout optimization across various geometry matrices (L: 10–150 μm, D: 5–50 μm). Successfully fabricated inverted-pyramid silicon molds via patterned AZ5214 lithography, ICP dry etching, and anisotropic KOH wet etching, enabling uniform texture replication via PDMS molding to optimize the light-harvesting capability of PSMs.

Work 4: Solar-driven Self-Powered Integrated Organic Electrochemical Transistor System for Sensing Applications

- **Highlights:** Engineered a self-powered, integrated biosensing system by coupling flexible perovskite solar cells (on PEI film) with organic electrochemical transistors (OECT) (on PI film). Led the microelectronic planar layout design and microfabrication of a rollable, flag-shaped flexible probe featuring gold interconnects on polyimide films, showcasing strong capabilities in low-power device integration and advanced packaging concepts.

PUBLICATIONS (Google Scholar Link)

Paper 1: Kang, B., & Yan, F. (2025). Emerging strategies for the large-scale fabrication of perovskite solar modules: from design to process. *Energy & Environmental Science*, 18(9), 3917-3954. (IF=35.2)

- **Highlights:** This work is an in-depth comprehensive review of the entire process of perovskite module device industrialization. This includes: **a)** the design of integrated perovskite solar devices in series; **b)** the methods of large-area semiconductor thin film deposition and fluid dynamic analysis; **c)** the devices performance enhancement of new materials; **d)** the operational stability

specifications.

Paper 2: Zhang, J., **Kang, B.**, & Yan, F. (2026). Molecular Design and Interfacial Functions of Self-Assembled Monolayers for Perovskite and Tandem Solar Cells. *Advanced Energy Materials*, e70979. (IF=26.8)

- *Highlights:* This review systematically describes the wide application of newly designed *P-type hole transport layers (self-assembled monolayers)* in perovskite solar cells, modules, and tandem solar cells. It primarily illustrated the high efficiency of these new p-type semiconductors in modulating interfacial energy levels and hole transport rates.

Paper 3: Zhuang, J., Liu, C., **Kang, B.**, Cheng, H., Xiao, M., Li, L., & Yan, F. (2024). Rapid surface reconstruction in air-processed perovskite solar cells by blade coating. *Advanced Materials*, 36(6), 2309869. (IF=28.9)

- *Highlights:* A newly designed molecule (DPGABr) with amino groups assists in the morphology reconstruction of large-area perovskite thin-film surfaces, with the surface *n-type character* enhancement.

HONORS AND AWARDS

Key Awards & Competitions

Gold Medal, 49th International Exhibition of Inventions Geneva 04/2024

- *Award-winning Innovation:* Flexible Perovskite Solar Modules Based on Surface Reconstruction Technology
- *Bringing the combined results of WORK1, 2, and 3—high-efficiency flexible PSMs—to the exhibition, received a gold medal from the judges. Following this, a consensus on follow-up cooperation was reached with a Saudi Arabian company at the exhibition.*

Third Prize, Formula Student China (FSC) 11/2020

- *Highlight:* Served as the Vice-group leader of the NPU team; participated in structural component design, compilation of the team marketing plan, and schedule *arrangement*.

Meritorious Winner (First Prize), Mathematical Contest in Modeling (MCM) 04/2020

- *Teamwork Wins:* Leading a three-person collaborative team to conduct mathematical modeling of the European fish industry and winning the gold medal.

University Scholarships & Honors

University-level Outstanding Student Scholarship 10/2020

College First Class Scholarship 10/2020

University-level Outstanding Student Cadre 05/2020

University-level Top 10 Social Practical Activity Pacesetter Scholarship 10/2019

INTERNSHIP EXPERIENCE

Quality Engineer, Kunshan Baojin Laser Tailor Welding Co., Ltd. 07/2020-08/2020

- Assisted in the preparation and distribution of on-site process documents

Lab Technician, SUNICHEM 07/2019-08/2019

- Responsible for work safety and environmental maintenance in lab

OVERSEAS EXCHANGE

The Belt and Road Initiative Exchange Program in the United Arab Emirates 01/2020

- Exchanged with the University of Sharjah and visited Sinopec Dubai

The Belt and Road Initiative Exchange Program in Vietnam 01/2019

- Cultural and economic exchange at Vietnam National University